astronomical objects situated at a variety of distances, we would have measured the universe's expansion rate at a variety of moments in the past. By comparing these expansion rates, we could determine how the expansion of space is slowing over time and thereby determine the deceleration parameter.

Carrying out this strategy for measuring the deceleration parameter thus requires two things: a means of determining the distance of a given astronomical object (so that we know how far back in time we are looking) and a means of determining the speed with which the object is receding from us (so that we know the rate of spatial expansion at that moment in the past). The latter ingredient is easier to come by. Just as the pitch of a police car's siren drops to lower tones as it rushes away from us, the frequency of vibration of the light emitted by an astronomical source also drops as the object rushes away. And since the light emitted by atoms like hydrogen, helium, and oxygen—atoms that are among the constituents of stars, quasars, and galaxies—has been carefully studied under laboratory conditions, a precise determination of the object's speed can be made by examining how the light we receive differs from that seen in the lab.

But the former ingredient, a method for determining precisely how far away an object is, has proven to be the astronomer's headache. The farther away something is, the dimmer you expect it to appear, but turning this simple observation into a quantitative measure is difficult. To judge the distance to an object by its apparent brightness, you need to know its intrinsic brightness—how bright it would be were it right next to you. And it is difficult to determine the intrinsic brightness of an object billions of light-years away. The general strategy is to seek a species of heavenly bodies that, for fundamental reasons of astrophysics, always burn with a standard, dependable brightness. If space were dotted with glowing 100-watt lightbulbs, that would do the trick, since we could easily determine a given bulb's distance on the basis of how dim it appears (although it would be a challenge to see 100-watt bulbs from significantly far away). But, as space isn't so endowed, what can play the role of standard-brightness lightbulbs, or, in astronomy-speak, what can play the role of *standard candles*? Through the years astronomers have studied a variety of possibilities, but the most successful candidate to date is a particular class of supernova explosions.